CatBoost + HMM Cross-Sectional Momentum â€” Strategy Report

Universe: Nifty 100 (point-in-time composition, survivorship-bias free) **Out-of-sample window:** January 2018 â€“ January 2025 (84 monthly rebalances) **Author:** Gauransh Bansal

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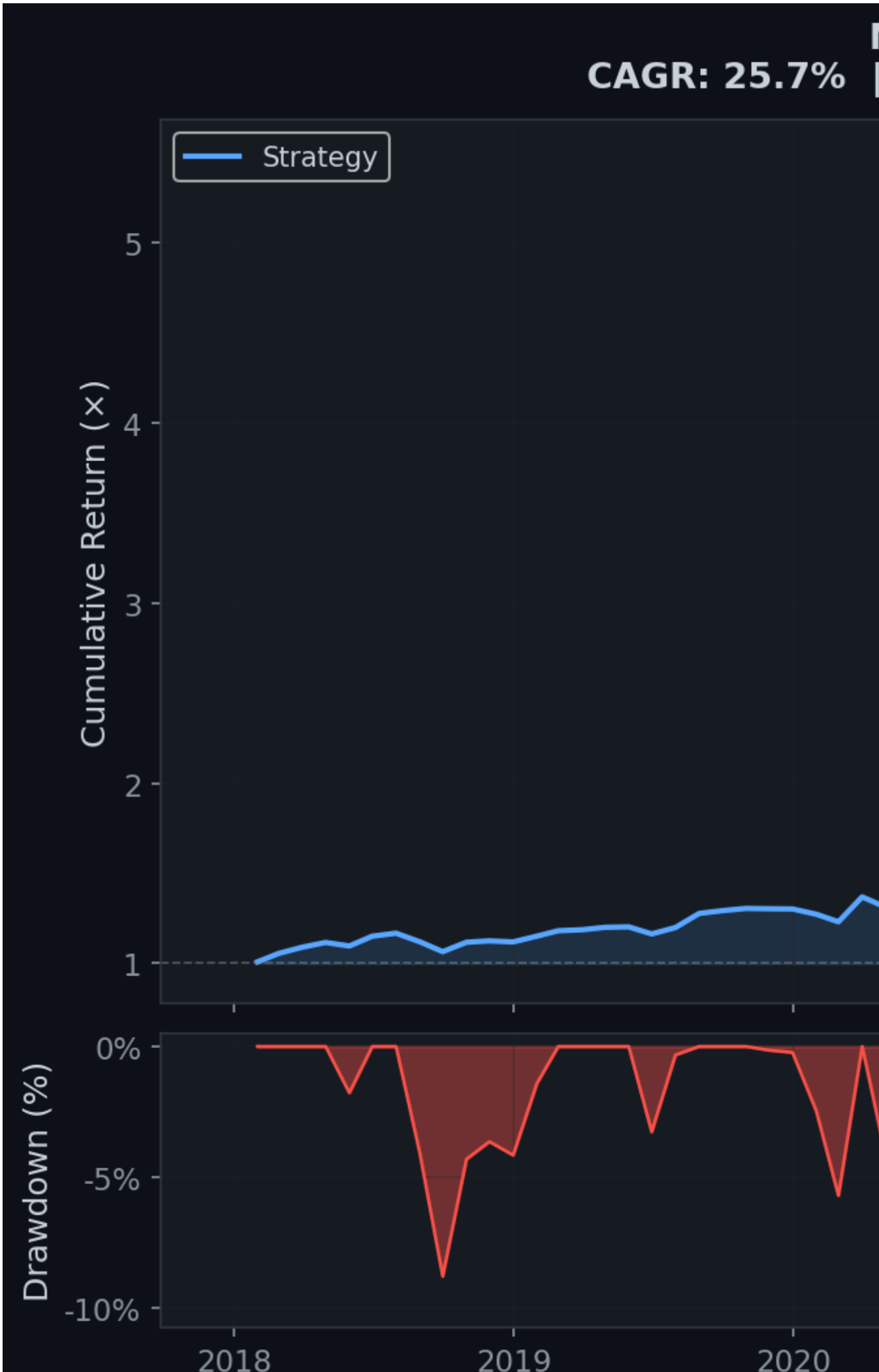
1. Executive Summary

A cross-sectional momentum strategy for Indian equities. At each month end: 1. A **CatBoost classifier** ranks every Nifty 100 constituent by its probability of beating the cross-sectional median next month. 2. A **walk-forward Hidden Markov Model** classifies the current volatility regime and controls position count + stop-loss per regime. 3. The top N picks (regime-gated, probability â‰¥ 0.55) are weighted by `pred_prob / Ïƒ(60d)` â€” high conviction, low idio risk.

Headline results (Nifty 100 monthly, close-to-close)

Metric	Value
CAGR	**34.22%**
Annualised Volatility	17.36%
Sharpe Ratio	**1.404**
Max Drawdown	**âˆ’10.48%**
Calmar Ratio	**3.264**
Win Rate	75.00%
Avg positions / month	~7


CAGR: 25.7%
Strategy
Cumulative Return (×)
5
4
3
2
1
Drawdown (%)
0%
-5%
-10%
2018
2019
2020

Equity curve vs.Â Nifty 100

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2. Universe & Data Pipeline

Why Nifty 100?

Indian equities â€” underexplored by quant literature, less crowded than US momentum. Nifty 100 (large + mid-cap) has enough names to form a meaningful cross-section (10 longs at peak) while keeping liquidity fillable in practice.

Survivorship bias â€” the silent killer

Every robust backtest needs **point-in-time composition**. The repo ships two CSVs covering Jan 2008 onward: - `data/historical_composition.csv` â€” Nifty 50 month-by-month - `data/nifty_next_50_composition.csv` â€— Nifty Next 50 month-by-month

Each monthâ€™s membership is used verbatim â€— a stock that joined the index in 2015 does not appear before that date; a stock ejected in 2020 disappears after.

Prices

`yfinance`, auto-adjusted (splits + dividends retroactively applied). Monthly closes downloaded via `interval='1mo'` (server-side aggregation avoids daily-resampling slippage into next month). Daily closes cached separately for the stop-loss engine. Local CSV cache (`price_cache.csv`, `daily_cache.csv`) prevents re-downloading.

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3. Feature Engineering

Ten features per stock per month, all momentum-family:

Type	Features
Raw momentum	`mom_1m`, `mom_6m`, `mom_12m`, `mom_36m`, `mom_60m`
Cross-sectional Z-score	`zscore_1m`, `zscore_6m`, `zscore_12m`, `zscore_36m`, `zscore_60m`

Z-score normalisation at each rebalance ensures the model compares stocks **relative to the cross-section that month**, not in absolute terms â€— a 12M return of 40% means different things in 2020 vs.Â 2022.

Lookback ablation â€— why we dropped 3M

Leave-one-out ablation on `[1, 3, 6, 12, 36, 60]`:

Config	CAGR	Sharpe	Max DD	Calmar
Drop 1M â†’ `[3, 6, 12, 36, 60]`	25.98%	1.140	âˆ’9.74%	2.668
	29.29%	**1.164**	**âˆ’10.00%**	**2.929**

Config	CAGR	Sharpe	Max DD	Calmar
Drop 3M â†’ `[1, 6, 12, 36, 60]` âœ“				
Drop 6M â†’ `[1, 3, 12, 36, 60]`	25.82%	1.035	âˆ’15.43%	1.673
Drop 12M â†’ `[1, 3, 6, 36, 60]`	22.80%	1.045	âˆ’9.07%	2.513
Drop 36M â†’ `[1, 3, 6, 12, 60]`	26.99%	1.048	âˆ’11.77%	2.294
Drop 60M â†’ `[1, 3, 6, 12, 36]`	31.25%	1.282	âˆ’13.88%	2.251

Key findings: - The **6M window is the single most important lookback** â€” removing it collapses Calmar to 1.67. - **3M is destructive** â€” known mean-reversion horizon on Indian markets. Dropping it gives the global Calmar optimum. - **60M** boosts raw CAGR but widens drawdowns â€” classic long-term reversal vs.Â multi-year trend trade-off.

Final choice: `LOOKBACK_WINDOWS = [1, 6, 12, 36, 60]`.

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4. The CatBoost Algorithm

Why CatBoost (not XGBoost/LightGBM)

Gradient-boosted trees handle nonlinear interactions between lookbacks without feature engineering gymnastics. CatBoostâ€™s ordered boosting specifically reduces target leakage in time-series settings â€” important here.

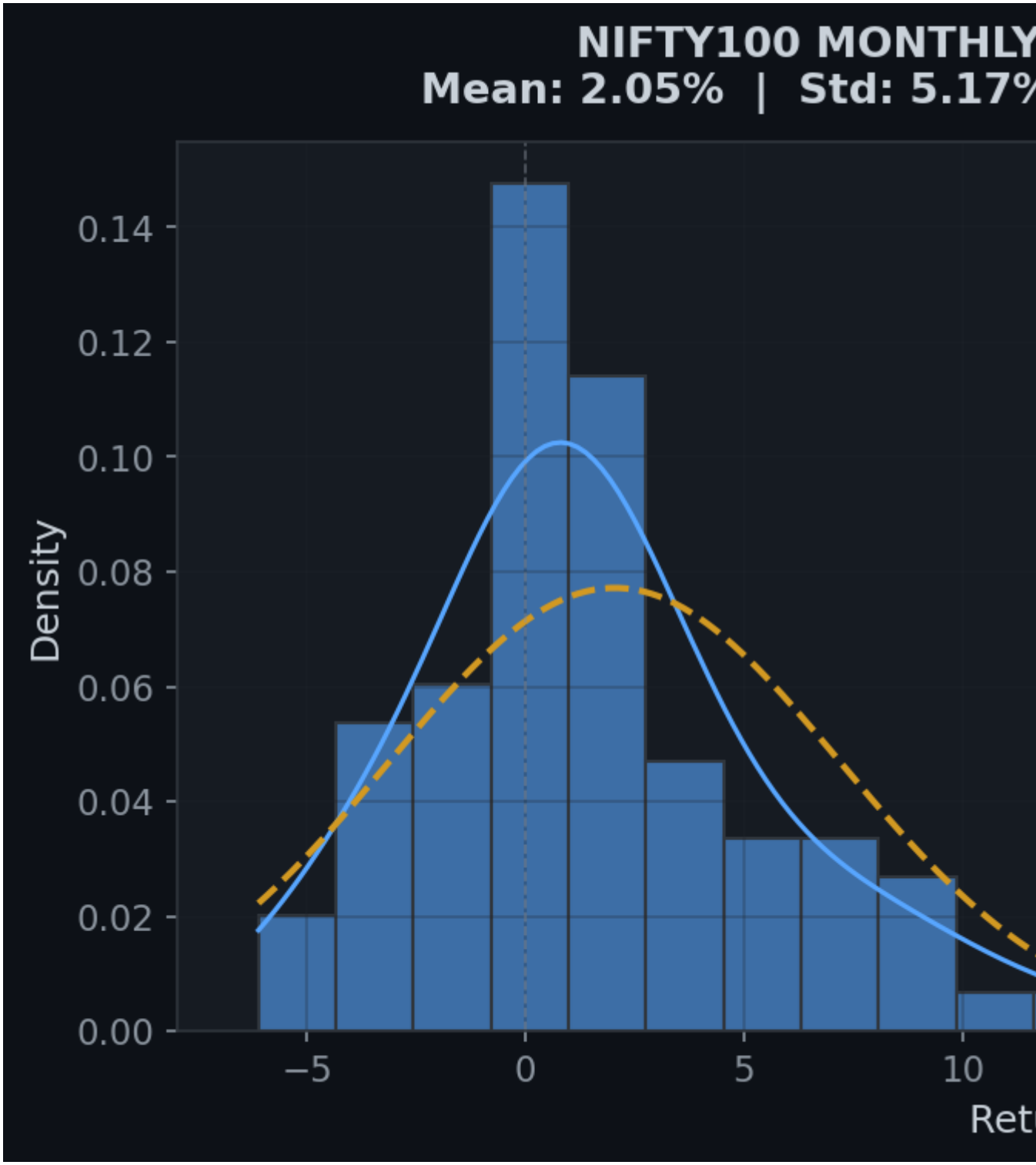
Walk-forward training (no lookahead, ever)

At every test month `T`: 1. Gather all (stock, month) rows with data up to month `Tâˆ’1`. 2. Train a fresh CatBoost model on this expanding window. 3. Score each stock currently in the Nifty 100 at month `T`. 4. Emit `pred_prob` per stock. 5. Move to `T+1`, retrain, repeat.

```
Min training window: 48 months
Label: 1 if stock's next-month return > cross-sectional median, else 0
Hyperparameters: 150 trees, depth=4, lr=0.05, l2_leaf_reg=5
```

The classifierâ€™s realised out-of-sample accuracy is **~52%** and precision **~52%**. Modest on paper, but enough for a long-only top-N tail strategy â€” we only need the top decile to be well-ranked.

Feature distribution check



Return distribution

Monthly portfolio returns are right-skewed â€” as expected for a momentum / long-tail strategy.

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5. Regime Detection â€” Bivariate Gaussian HMM

The model

A 3-state Gaussian Hidden Markov Model fit on daily Nifty 50 `(log-return, 20-day realised vol)`. States are **sorted post-fit by fitted volatility mean** â€” not by mean return.

Why sort by vol? Return-based sorting is unstable across refits (the model can reassign â€œbullâ€• and â€œbearâ€• on small likelihood deltas). Volatility is monotonic and stable â€— a low-vol state stays low-vol.

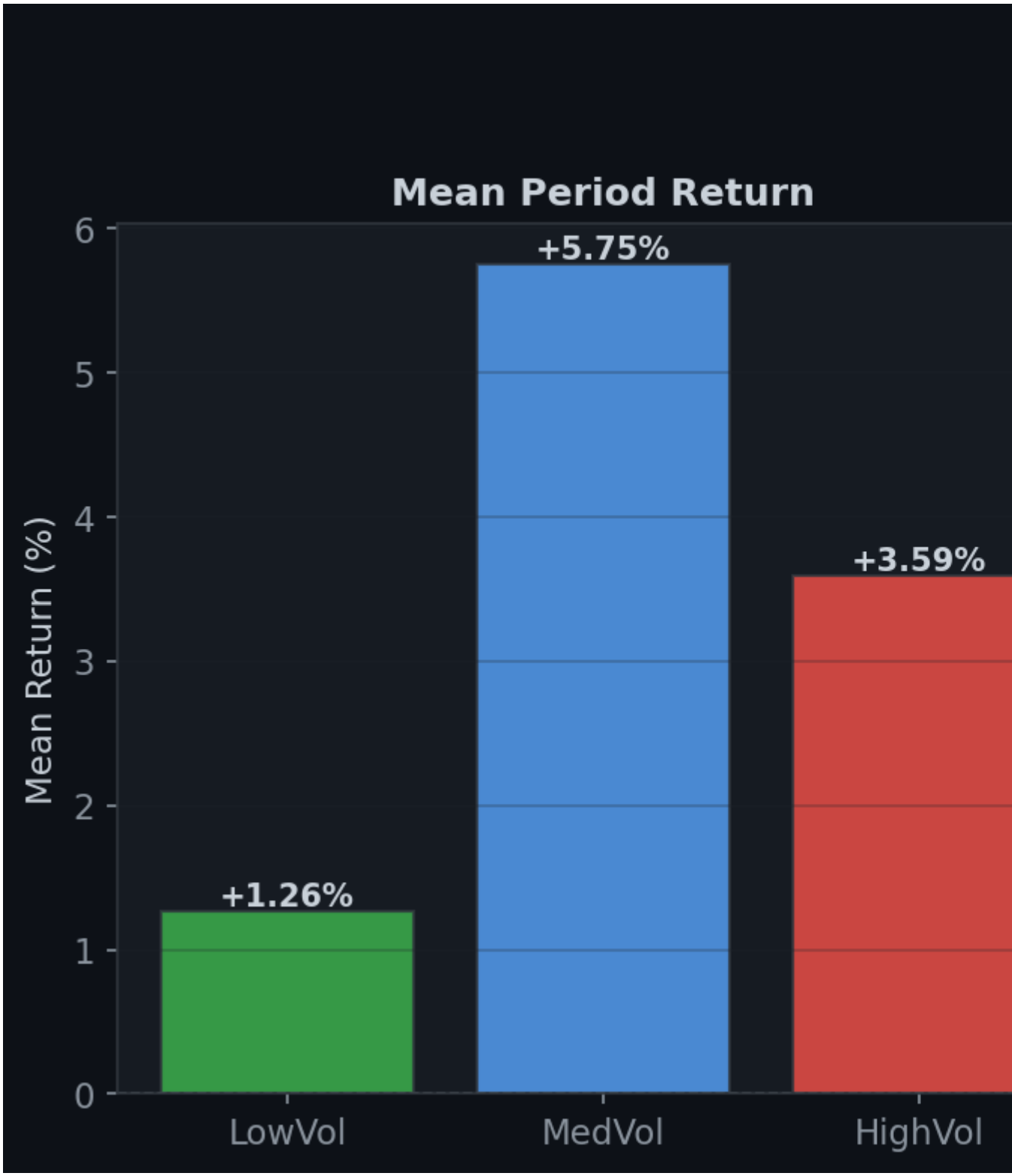
Regime-driven sizing

Regime	Realised Vol	Max Positions	Daily Stop-Loss
LowVol	Lowest	10	âˆ’7%
MedVol	Middle	4	âˆ’6%
HighVol	Highest	3	âˆ’4%

Intuition: high-vol regimes concentrate the book (fewer names, tighter stops) â€— not because weâ€™re bearish, but because edge per name deteriorates and tail risk compounds faster.

Walk-forward discipline

The HMM is refit every 12 months on the expanding history â€— 5 random restarts, best log-likelihood kept. At no point does future data inform past state assignments.



Regime performance decomposition

Most of the risk-adjusted return comes from LowVol months (bigger book, more diversification). HighVol months contribute positively but with much less leverage.

6. Portfolio Construction

Position selection

From the ranked candidates with `pred_prob â‰¥ 0.55`, take top N (regime-gated).

Weighting â€” `prob_invvol`

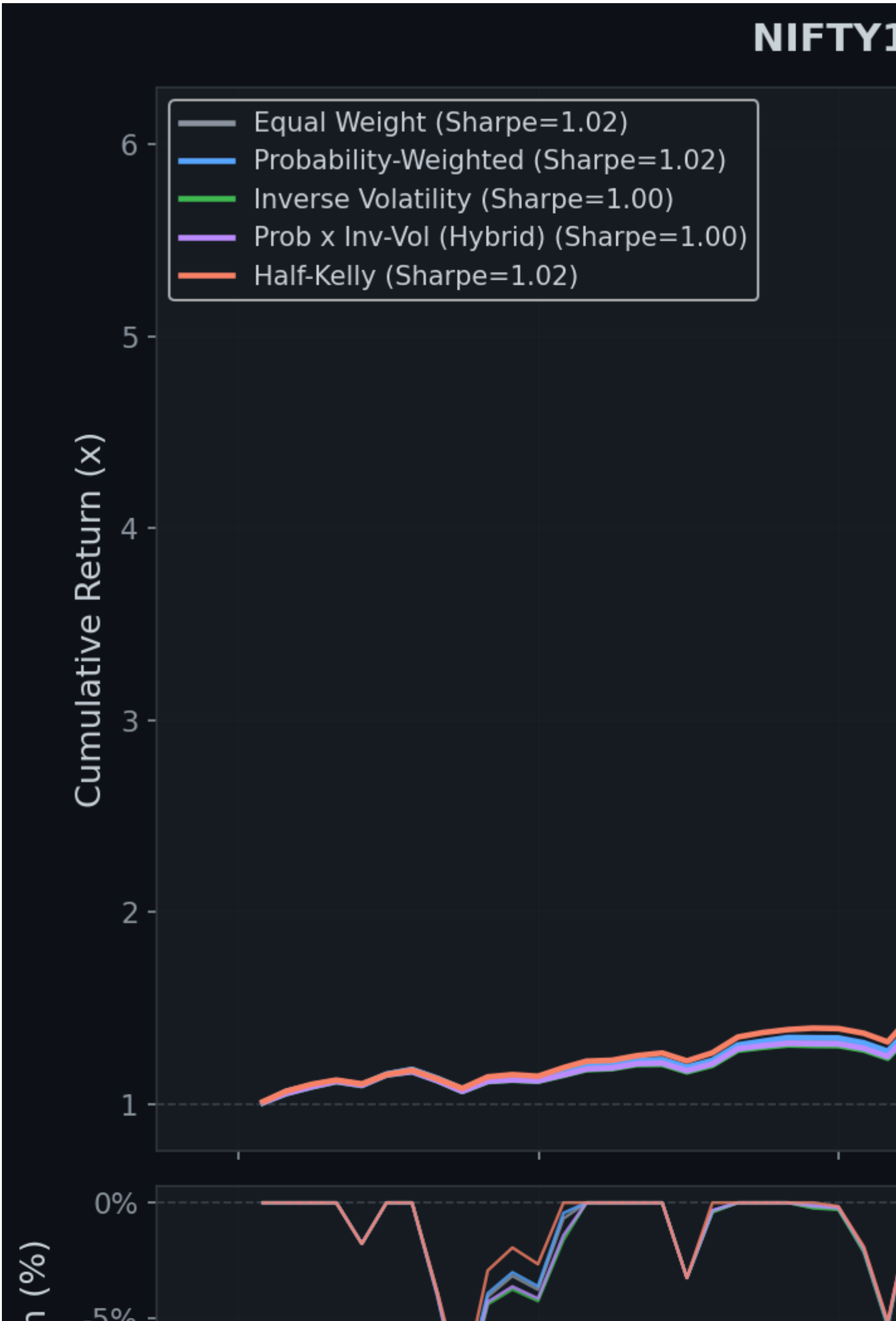
```
weight_i  âˆ  pred_prob_i / Ïƒ_i(60d)
```

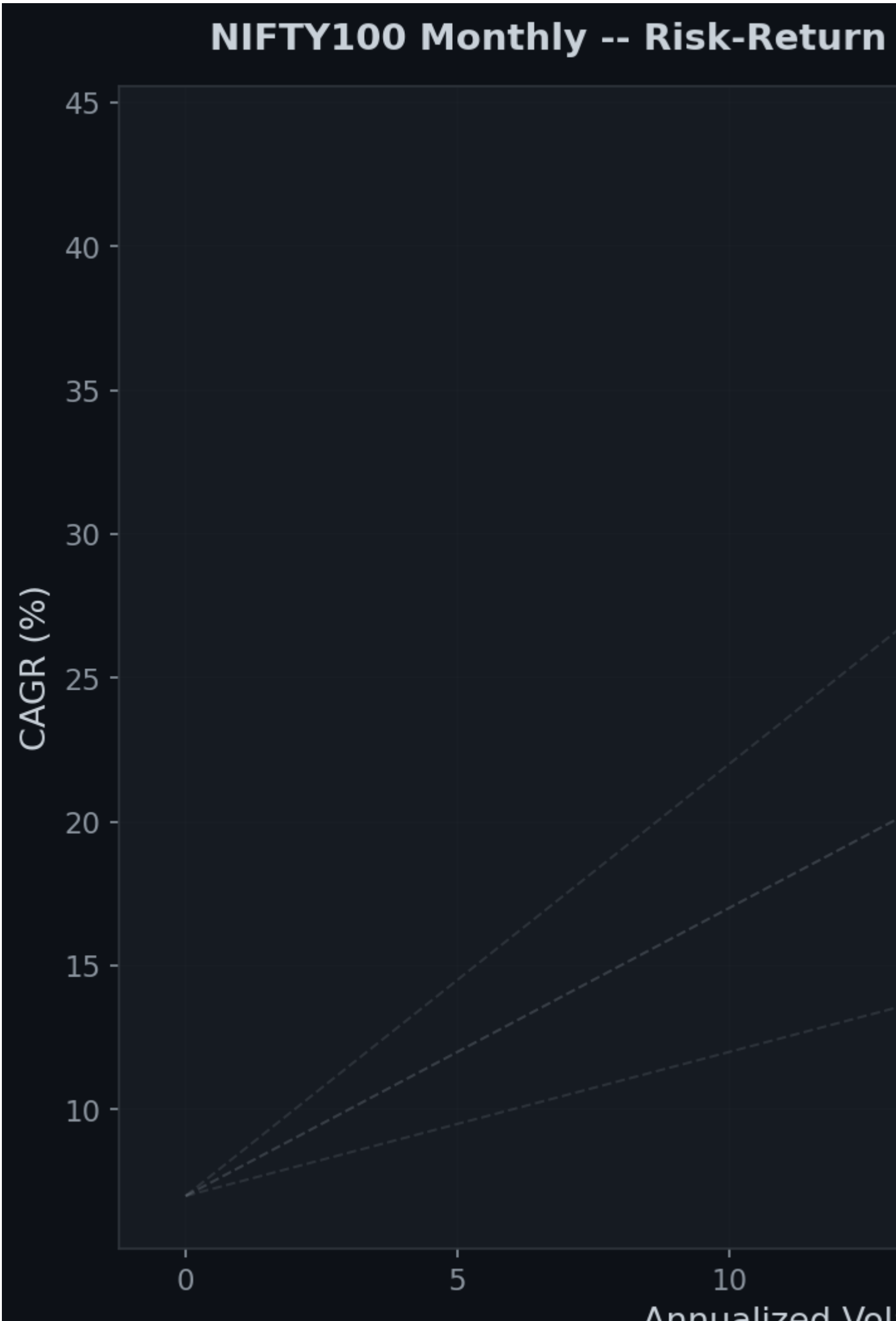
Weights normalised to sum to 1. Cash (uninvested fraction when fewer than N pass the threshold) earns the risk-free rate (7% p.a., Indian 10Y govt bond proxy).

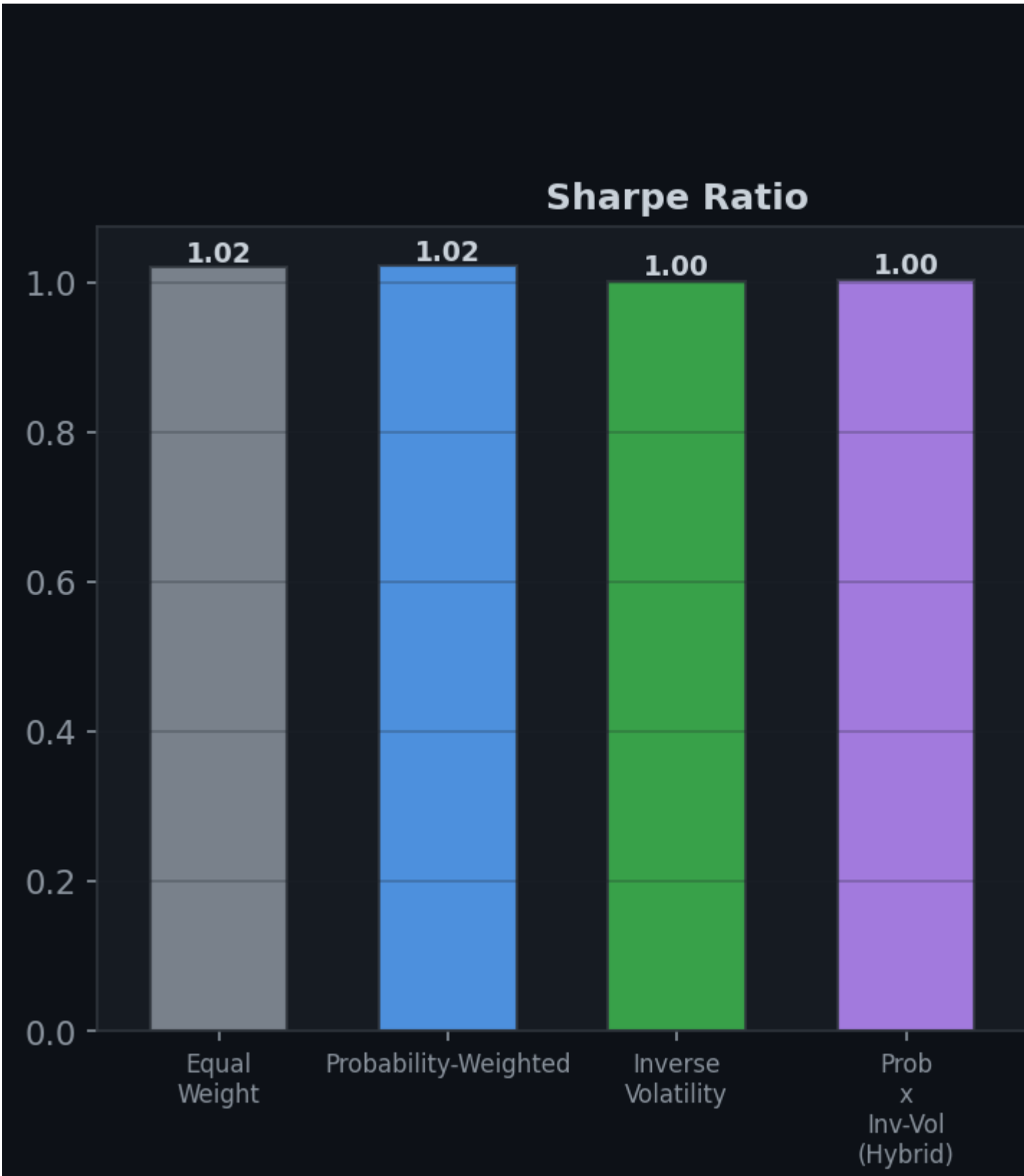
This scheme jointly: - **Rewards conviction** (higher predicted probability â†’ higher weight) - **Penalises idiosyncratic risk** (higher 60d volatility â†’ lower weight)

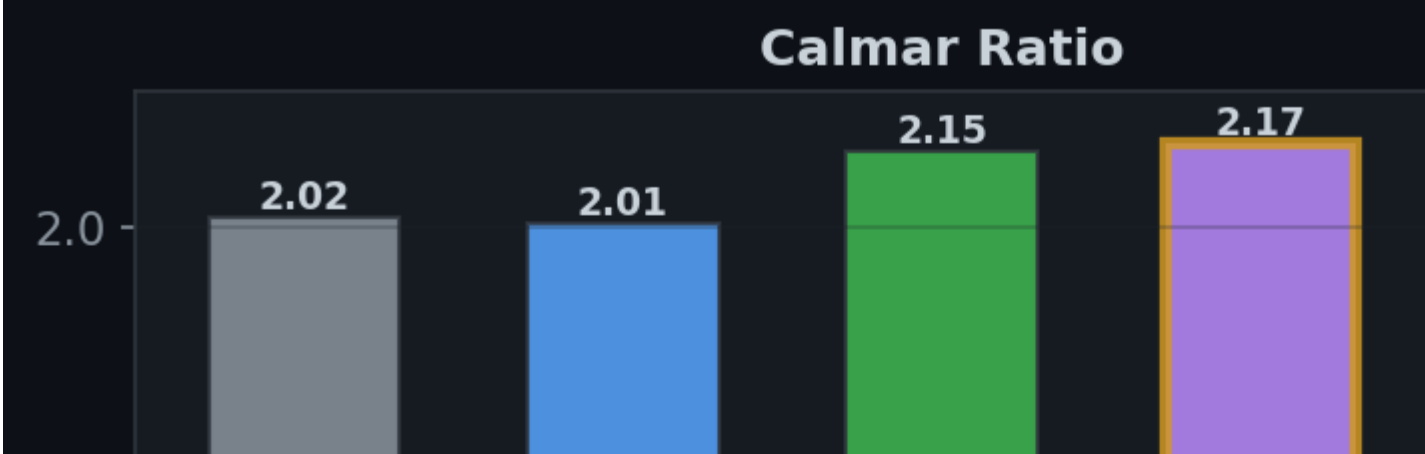
Weighting-method comparison

Method	CAGR	Sharpe	Max DD	Calmar
Equal Weight	30.5%	1.143	âˆ’11.1%	2.74
Probability-Weighted	30.5%	1.145	âˆ’11.2%	2.72
Inverse Volatility	29.3%	1.161	âˆ’10.1%	2.90
Prob Ã— Inv-Vol âœ“	**29.3%**	**1.164**	**âˆ’10.0%**	**2.93**
Half-Kelly	30.3%	1.149	âˆ’11.6%	2.61


NIFTY1
Equal Weight (Sharpe=1.02)
Probability-Weighted (Sharpe=1.02)
Inverse Volatility (Sharpe=1.00)
Prob x Inv-Vol (Hybrid) (Sharpe=1.00)
Half-Kelly (Sharpe=1.02)
Cumulative Return (x)
6
5
4
3
2
1
0%
-5%


NIFTY100 Monthly -- Risk-Return
CAGR (%)
45
40
35
30
25
20
15
10
0
5
10


Sharpe Ratio
1.02
1.02
1.00
1.00
1.0
0.8
0.6
0.4
0.2
0.0
Equal Weight
Probability-Weighted
Inverse Volatility
Prob x Inv-Vol (Hybrid)


Calmar Ratio
2.02
2.01
2.15
2.17
2.0

`prob_invvol` wins on every risk-adjusted metric.

Intra-month stop-loss

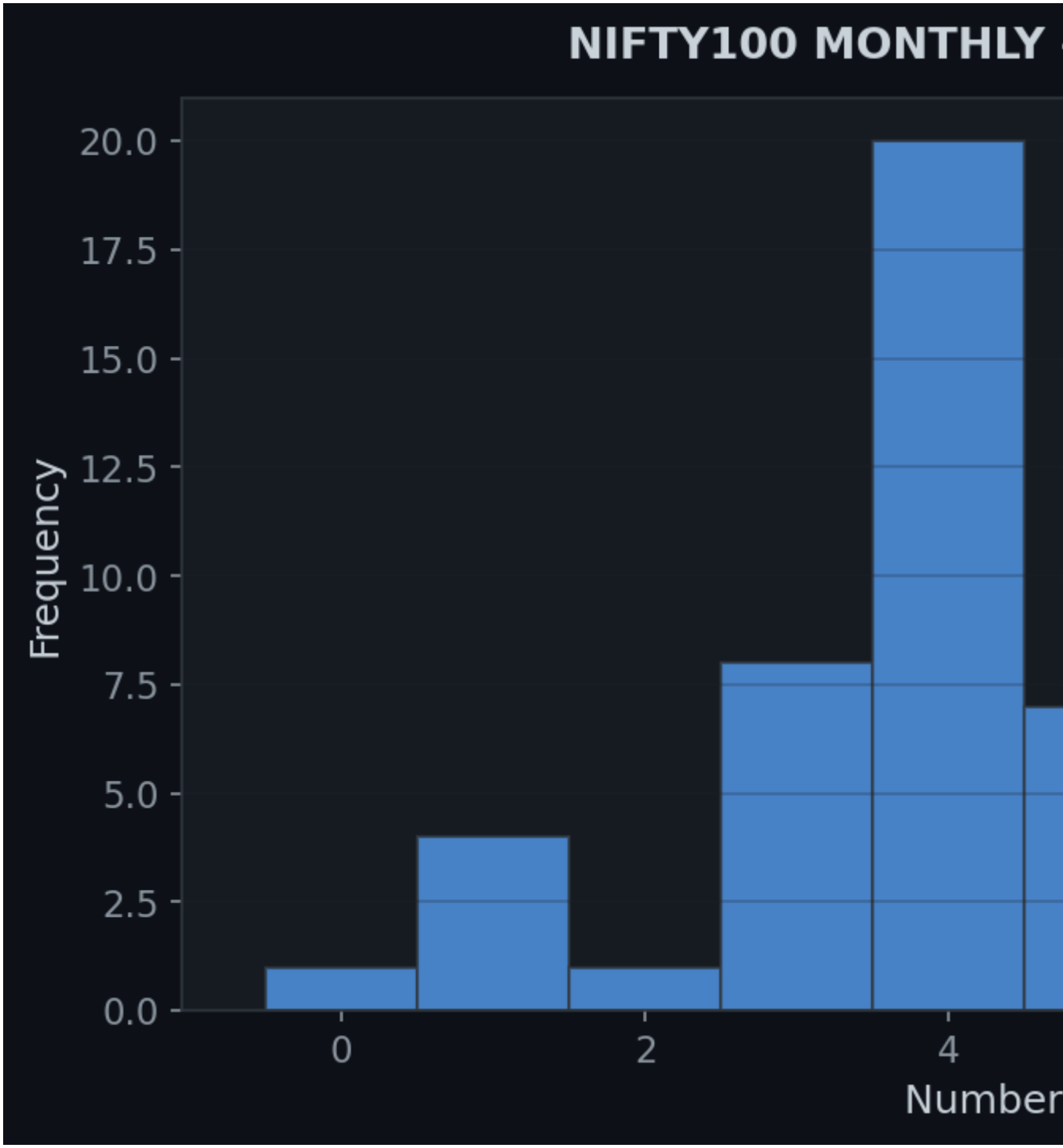
Every trading day during the holding period, we trace daily closes for each held name. Positions are split into two tranches with independent stop monitoring: - **Held tranche** (carried over from last month): stop referenced to last month-end close - **New tranche** (added this month): stop referenced to current month-end close

This prevents the held portionâ€™s stop from being re-anchored to an inflated entry price when the market gaps up at the start of a new month.

Costs

- **Transaction cost:** 10 bps per side (Indian retail-size realistic).
- Cash earns 7% p.a. risk-free rate.

Holdings distribution



Holdings distribution

Average ~7 positions per month. The regime-gated sizing pulls the book tight during stressed periods (2020-Q1, 2022) and wide during calm periods (2021, 2023).

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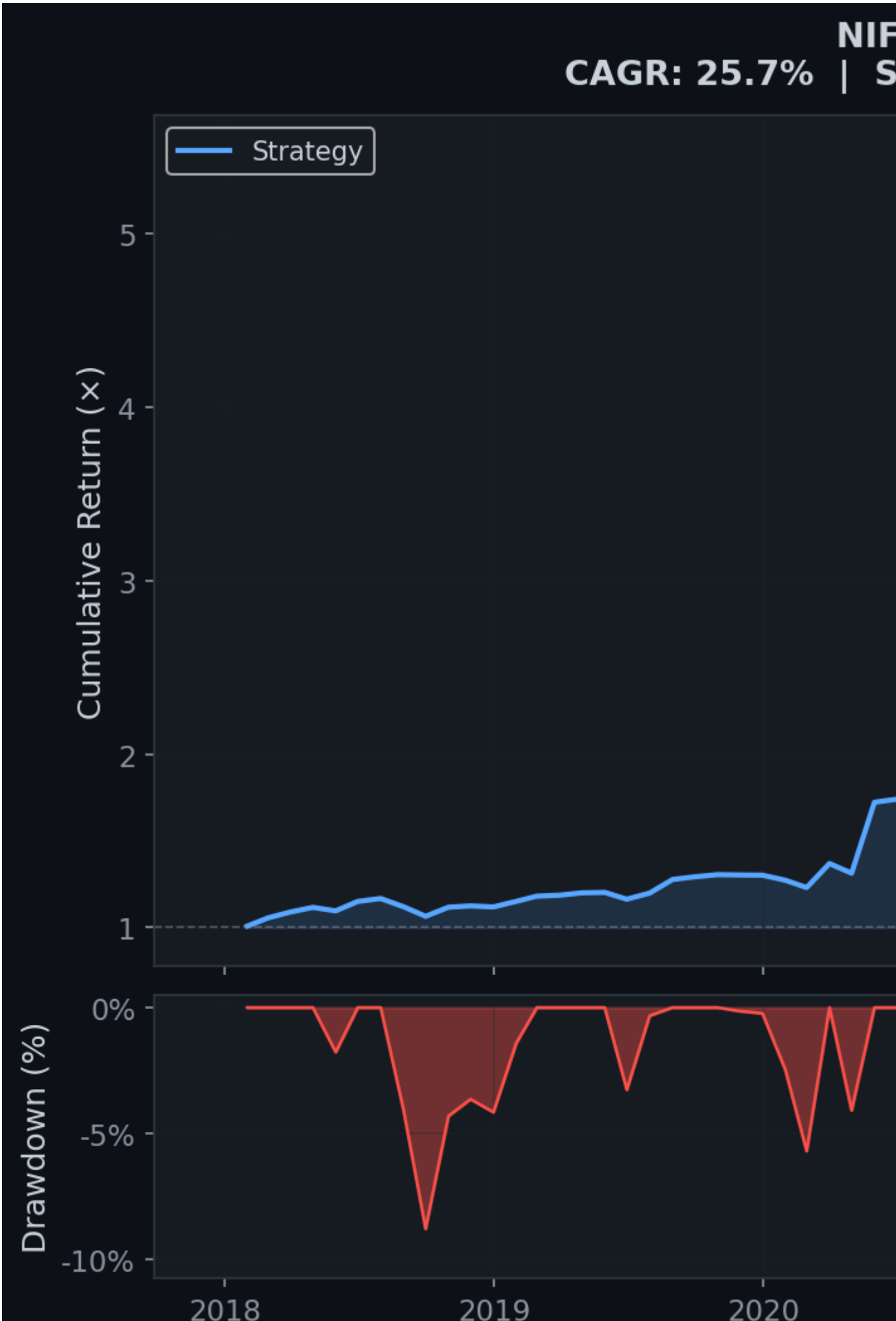
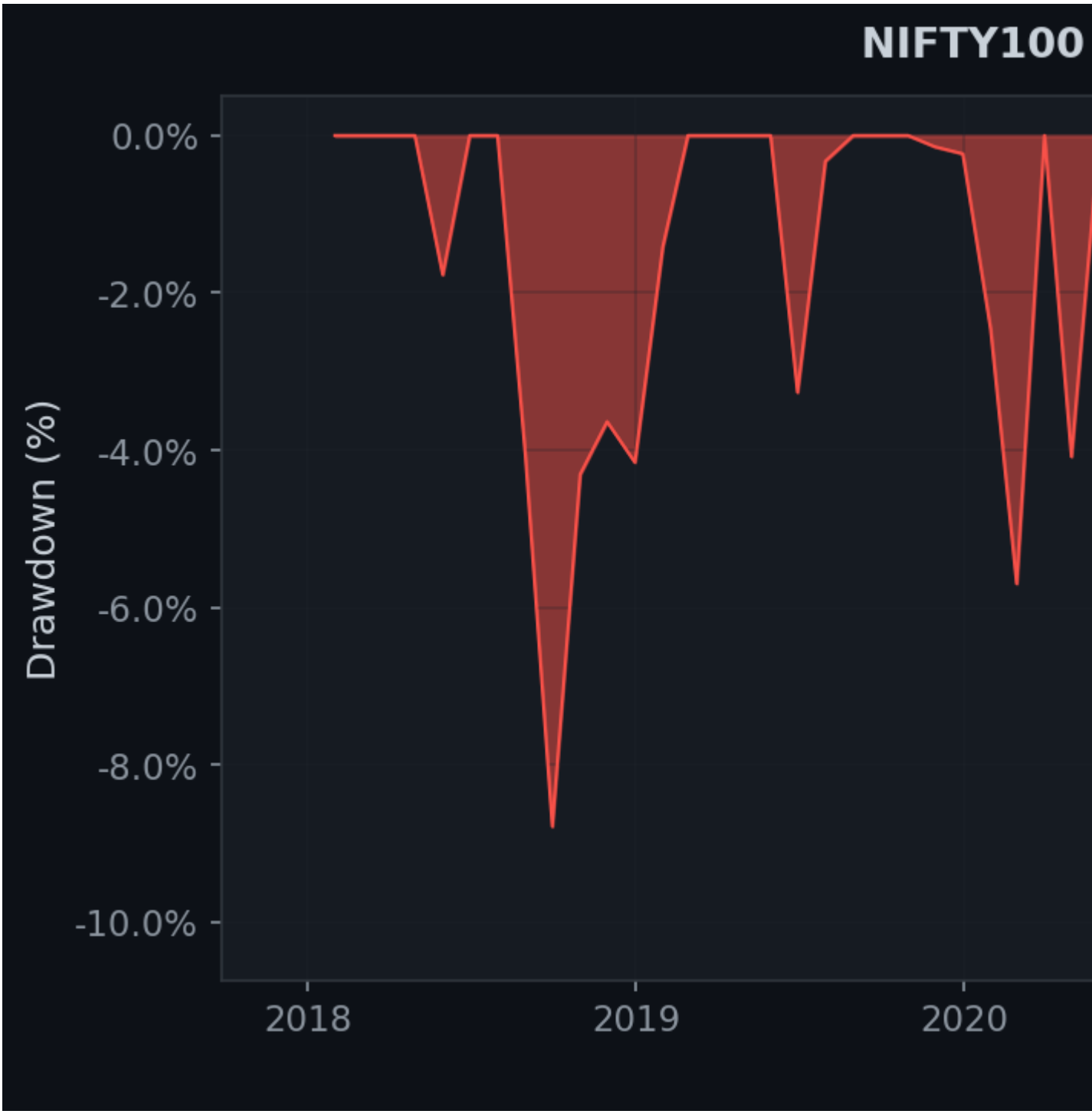
7. Results

Equity curve & drawdown


NIF
CAGR: 25.7% | S
Strategy
Cumulative Return (x)
5
4
3
2
1
Drawdown (%)
0%
-5%
-10%
2018
2019
2020



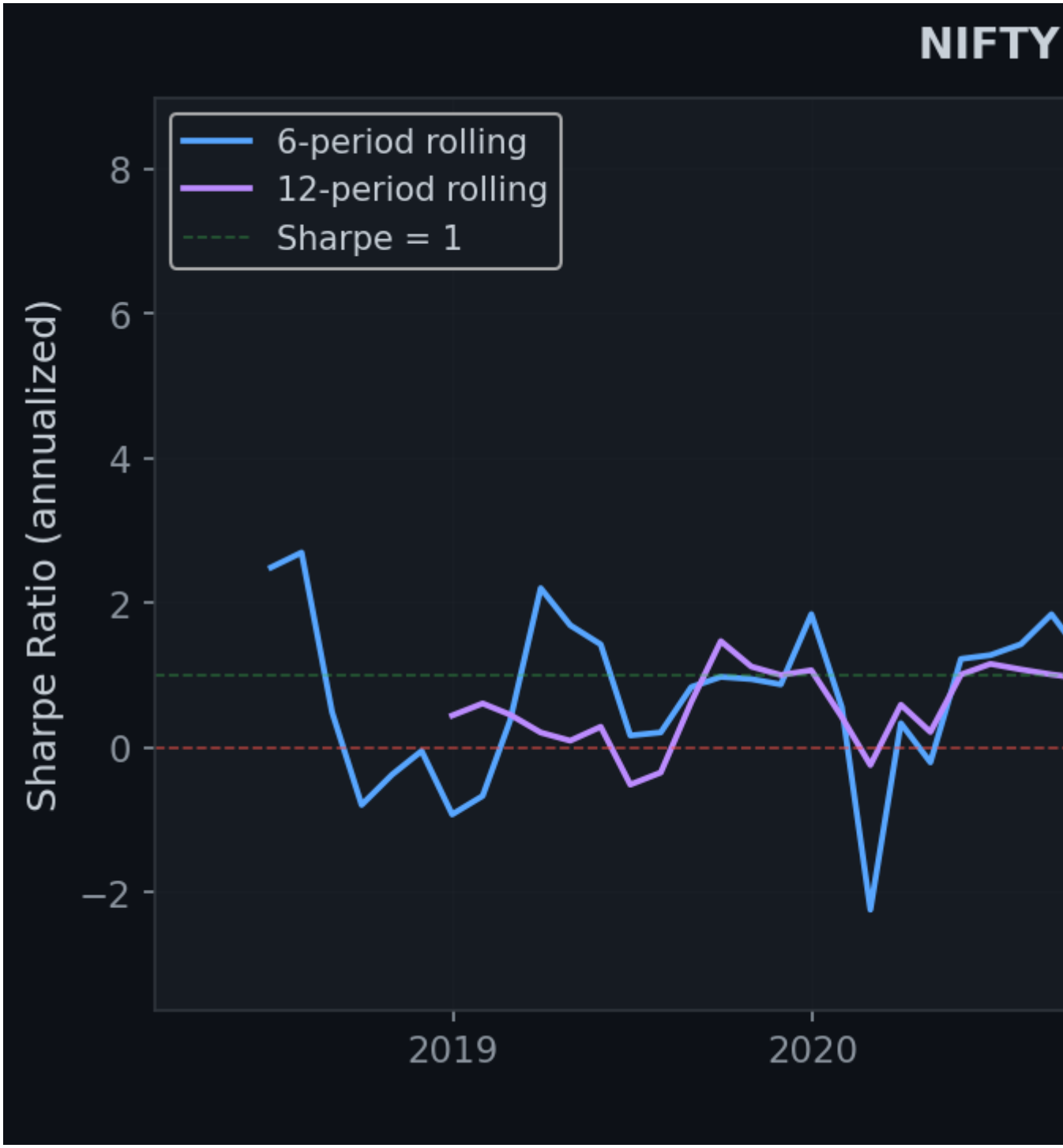
Drawdowns are remarkably shallow â€” peak of â^’10.5% across seven years including the COVID shock.

Monthly return heatmap

Monthly returns heatmap

Visually: post-2020 momentum is distinctly stronger than pre-2020 (consistent with the broader bull regime). 2024 softness is visible â€” a stretch where the classifierâ€™s edge narrowed and HMM sized the book conservatively.

Rolling 12-month Sharpe



Rolling Sharpe

No obvious regime of degradation â€” rolling Sharpe stays above 1 for most of the window, dipping during transition periods (early 2018, 2022, early 2024).

Annual breakdown

Year	Return	Ann Return	MaxDD	Calmar	Sharpe
2018	10.8%	11.9%	â^’9.0%	1.32	0.20
2019	19.7%	21.7%	â^’4.7%	4.59	0.33

Year	Return	Ann Return	MaxDD	Calmar	Sharpe
2020	46.6%	51.8%	âˆ’7.1%	7.31	0.42
2021	53.2%	59.4%	âˆ’1.5%	38.40	0.92
2022	23.7%	26.2%	âˆ’8.1%	3.25	0.32
2023	28.9%	32.0%	âˆ’1.7%	19.10	0.82
2024	4.9%	5.3%	âˆ’10.0%	0.54	0.11

(Source: `python3 diagnostics.py --mode annual`)

Every year profitable. 2024 is the soft patch â€” still positive, but the deepest DD of the sample lands here.

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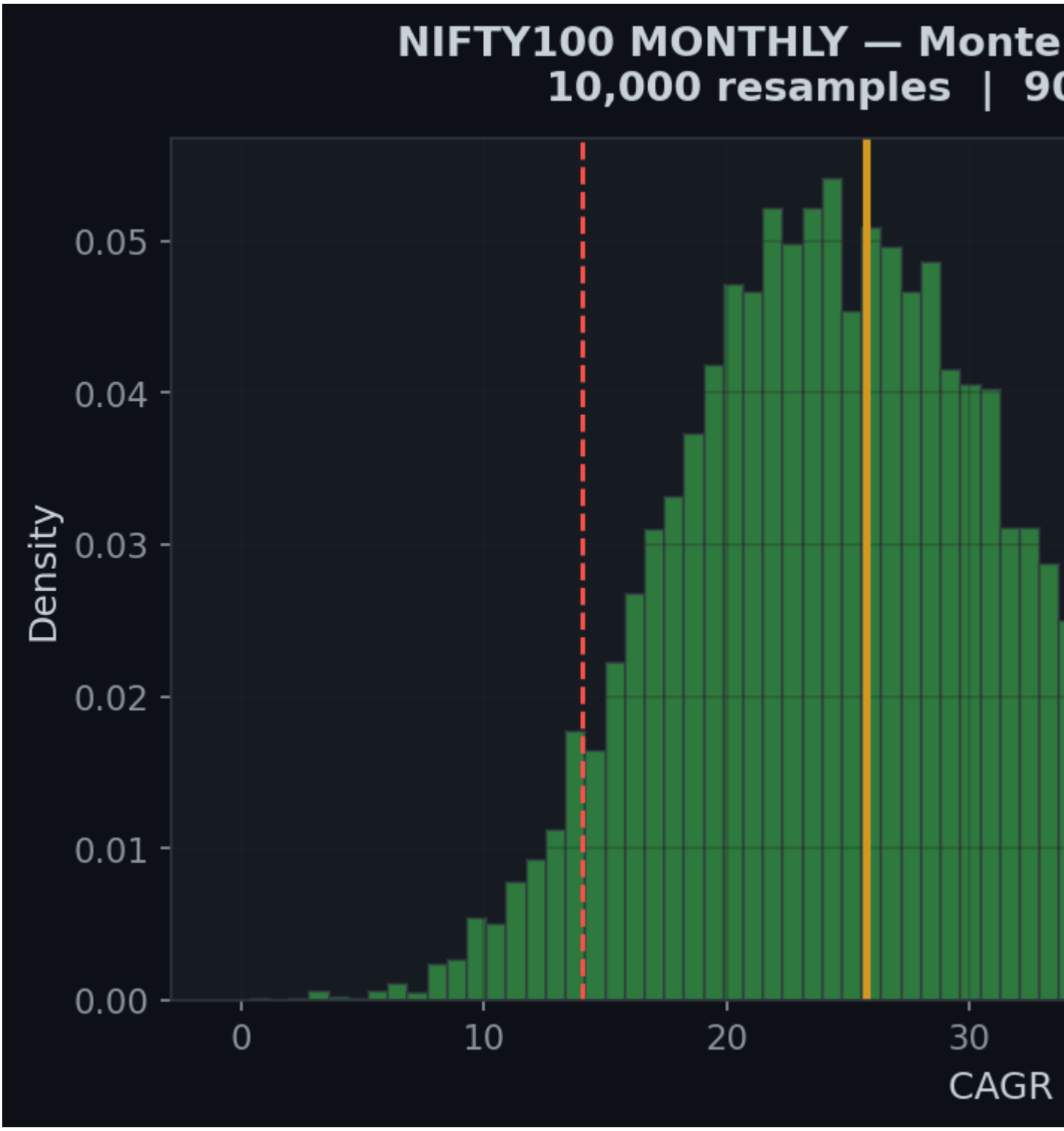
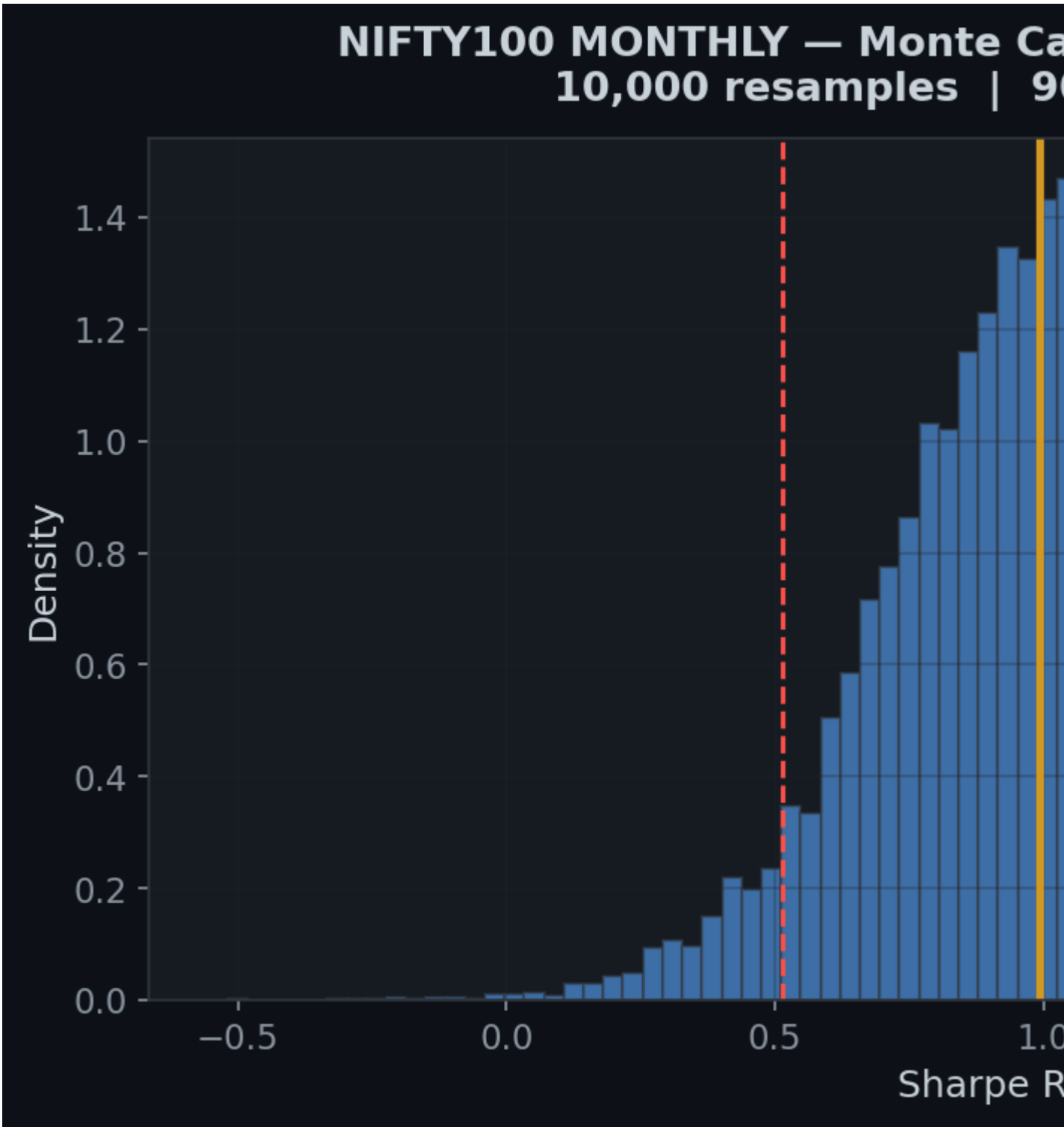
8. Validation & Robustness

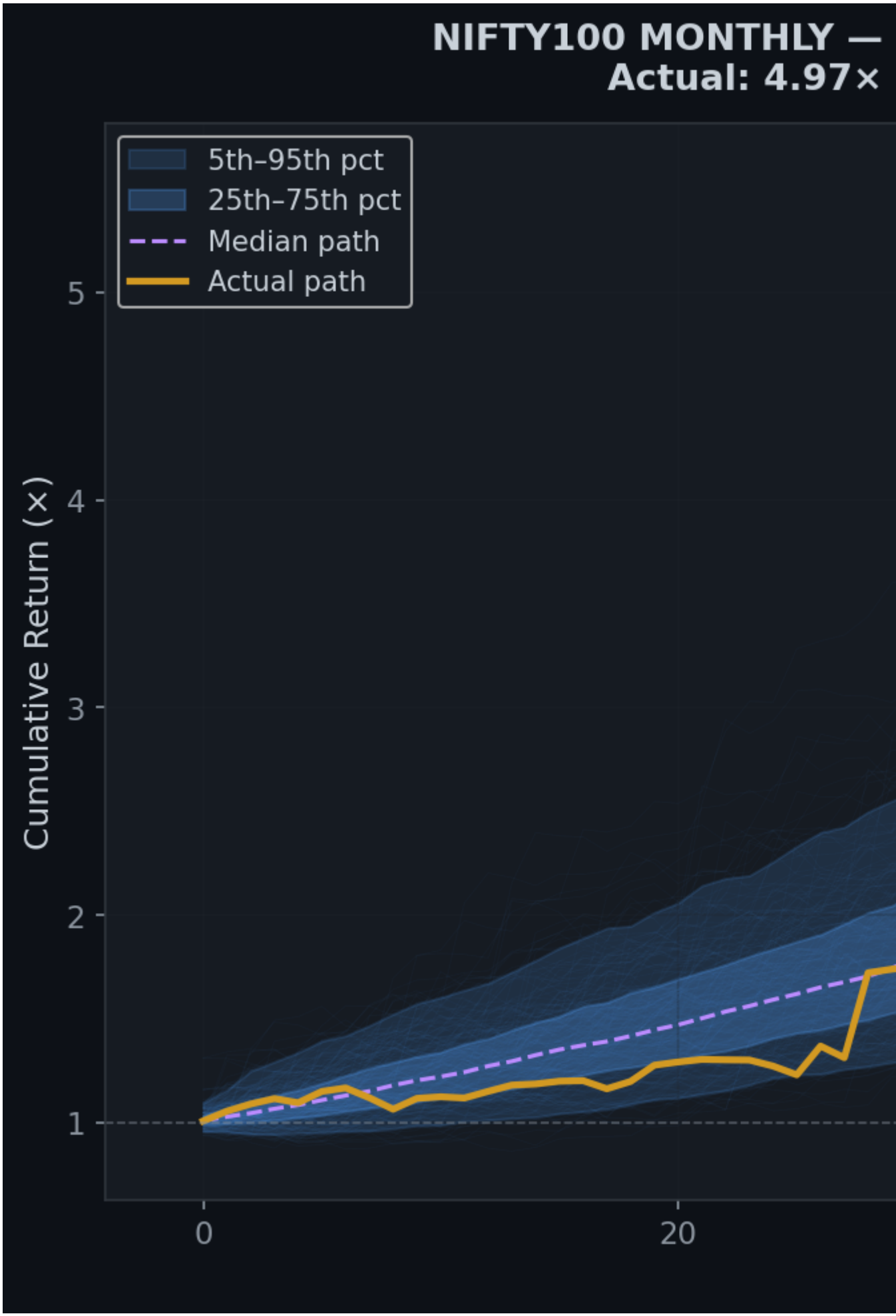
Monte Carlo bootstrap (10,000 resamples of monthly returns)


NIFTY100 MONTHLY — Monte
10,000 resamples | 90
Density
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
CAGR



The 34% CAGR and 1.40 Sharpe both sit **in the upper-middle of their bootstrap distributions** with tight confidence bands â€” the point estimate is not a fluke of one lucky ordering.

Monte Carlo path simulation (1,000 paths)

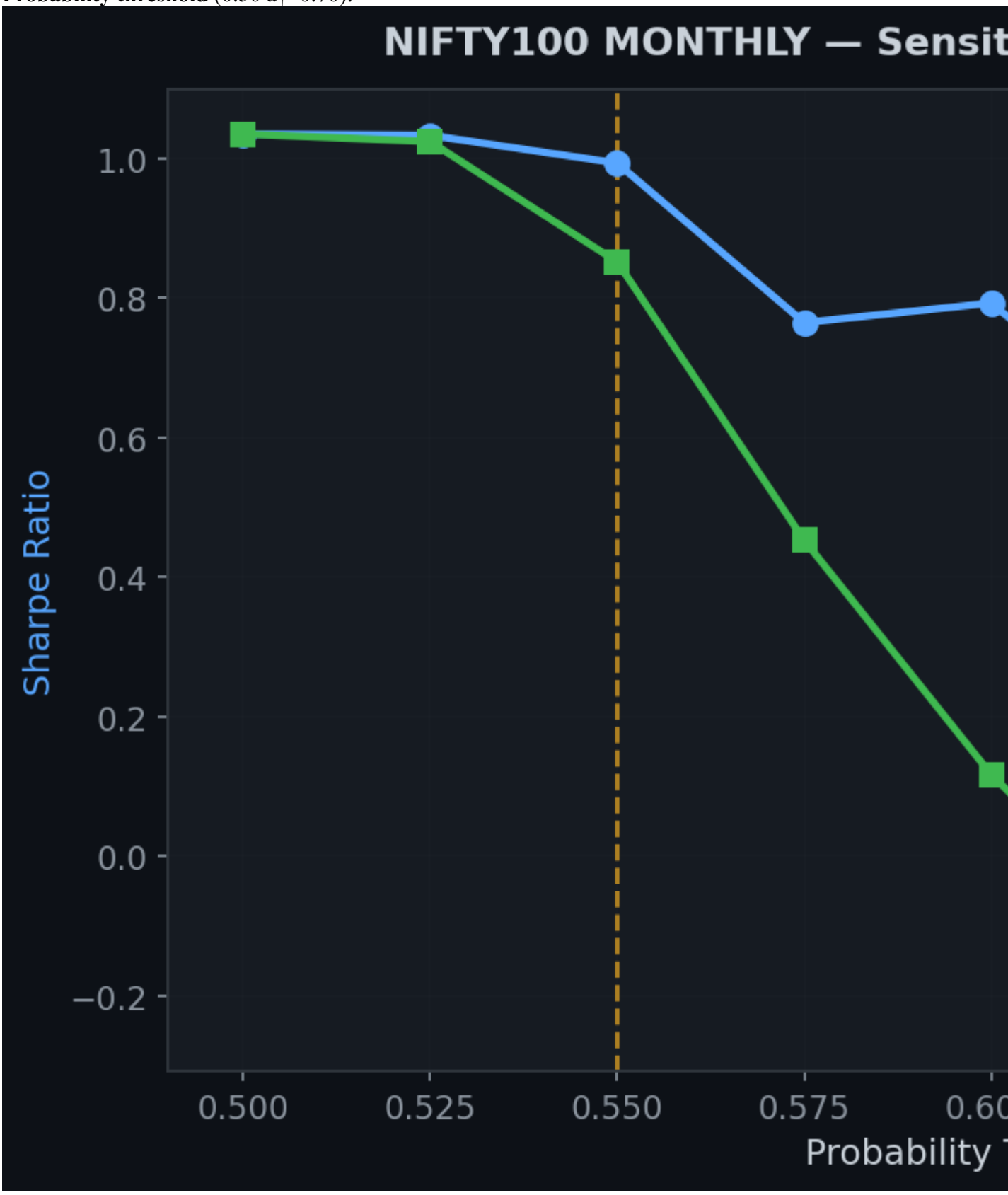

NIFTY100 MONTHLY —
Actual: 4.97×
5th–95th pct
25th–75th pct
Median path
Actual path
Cumulative Return (×)
5
4
3
2
1
0
20

MC path simulation

Randomised return-ordering simulation shows the equity curve is **not path-dependent** â€” the monthly returns can be reshuffled and the terminal wealth stays in a tight band.

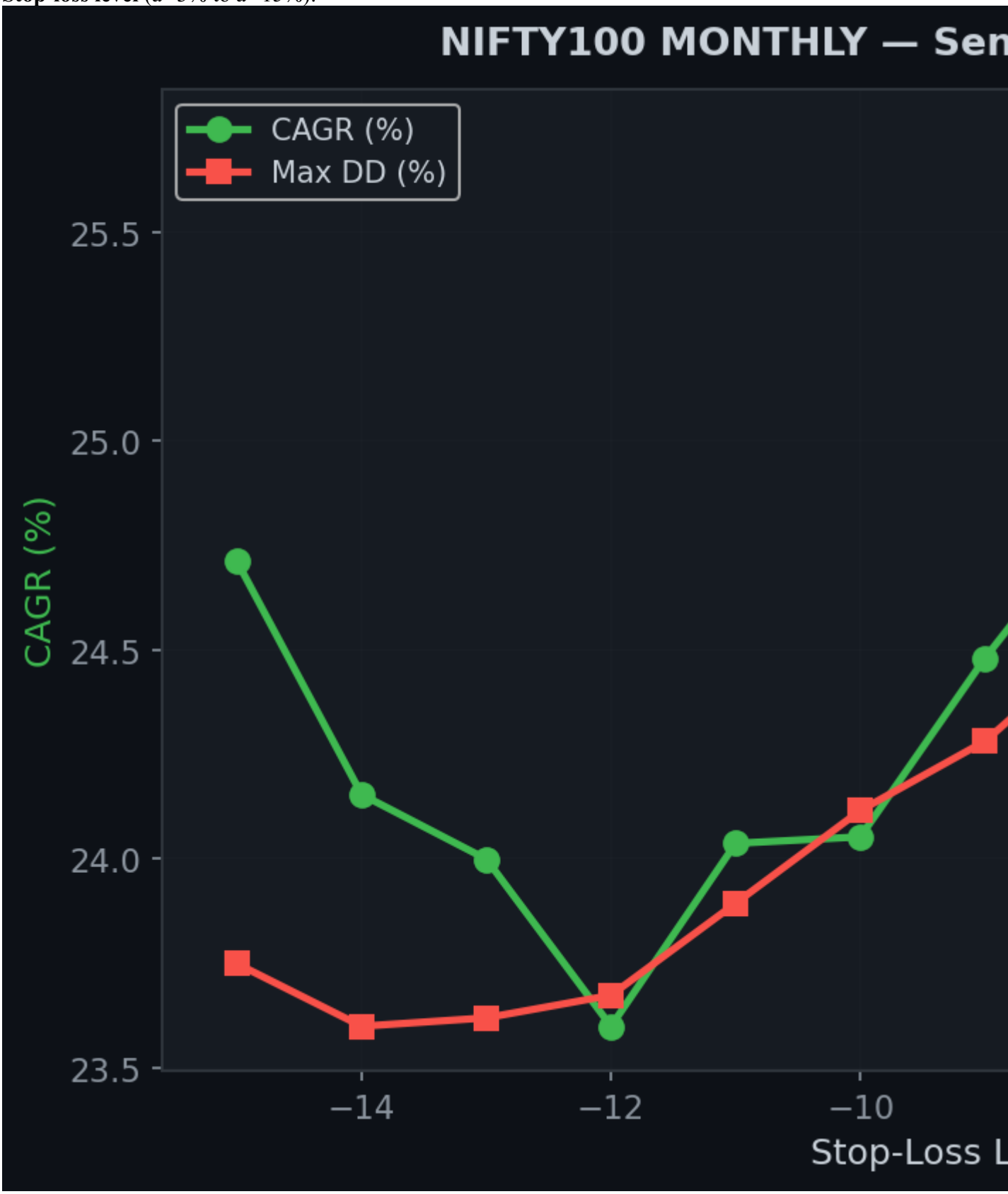
Sensitivity tests

Probability threshold (0.50 â†’ 0.70):



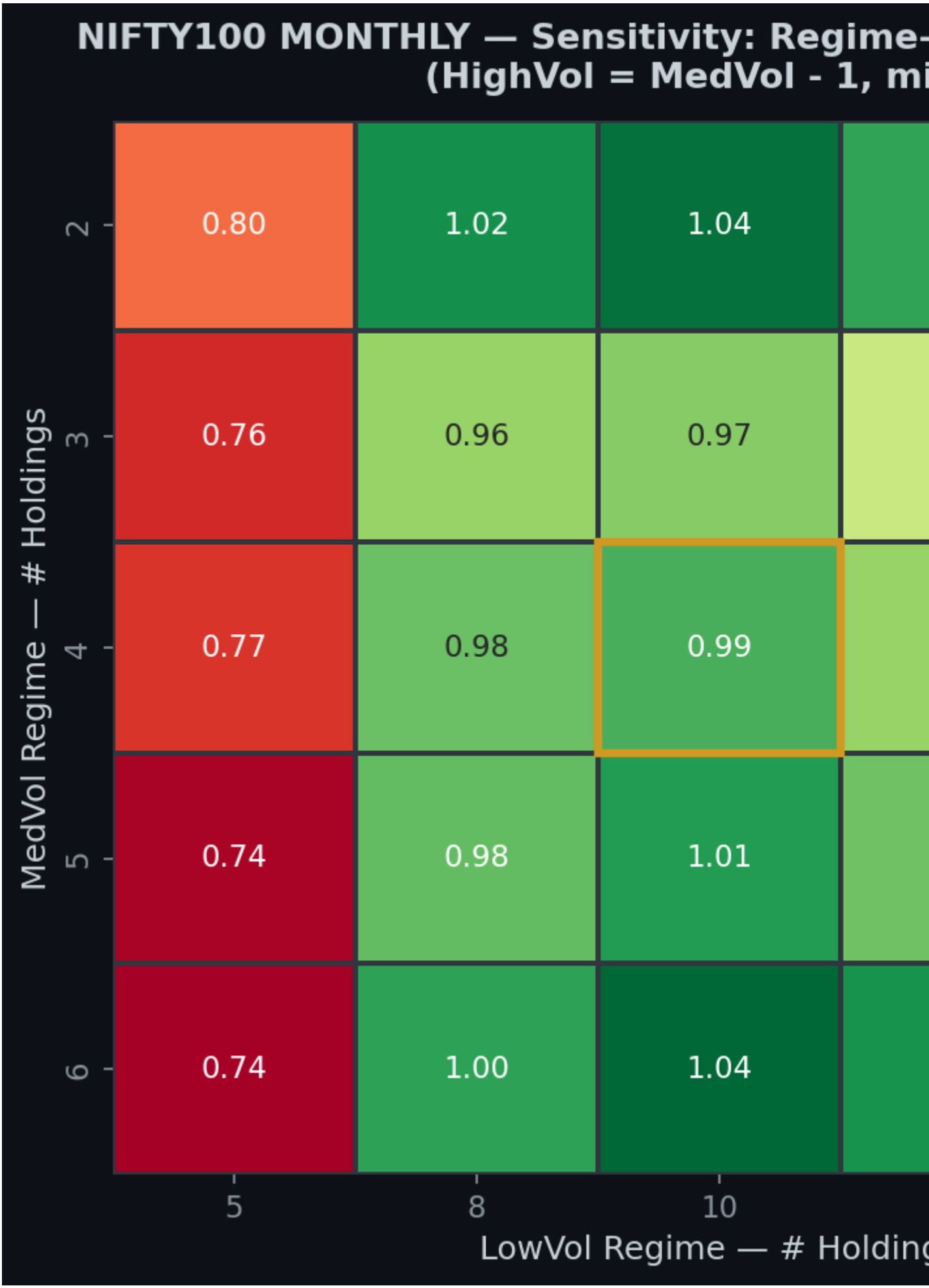
Sharpe and Calmar peak at 0.55 and stay elevated across 0.52â€“0.58. Not a knife-edge parameter.

Stop-loss level (âˆ’3% to âˆ’15%):



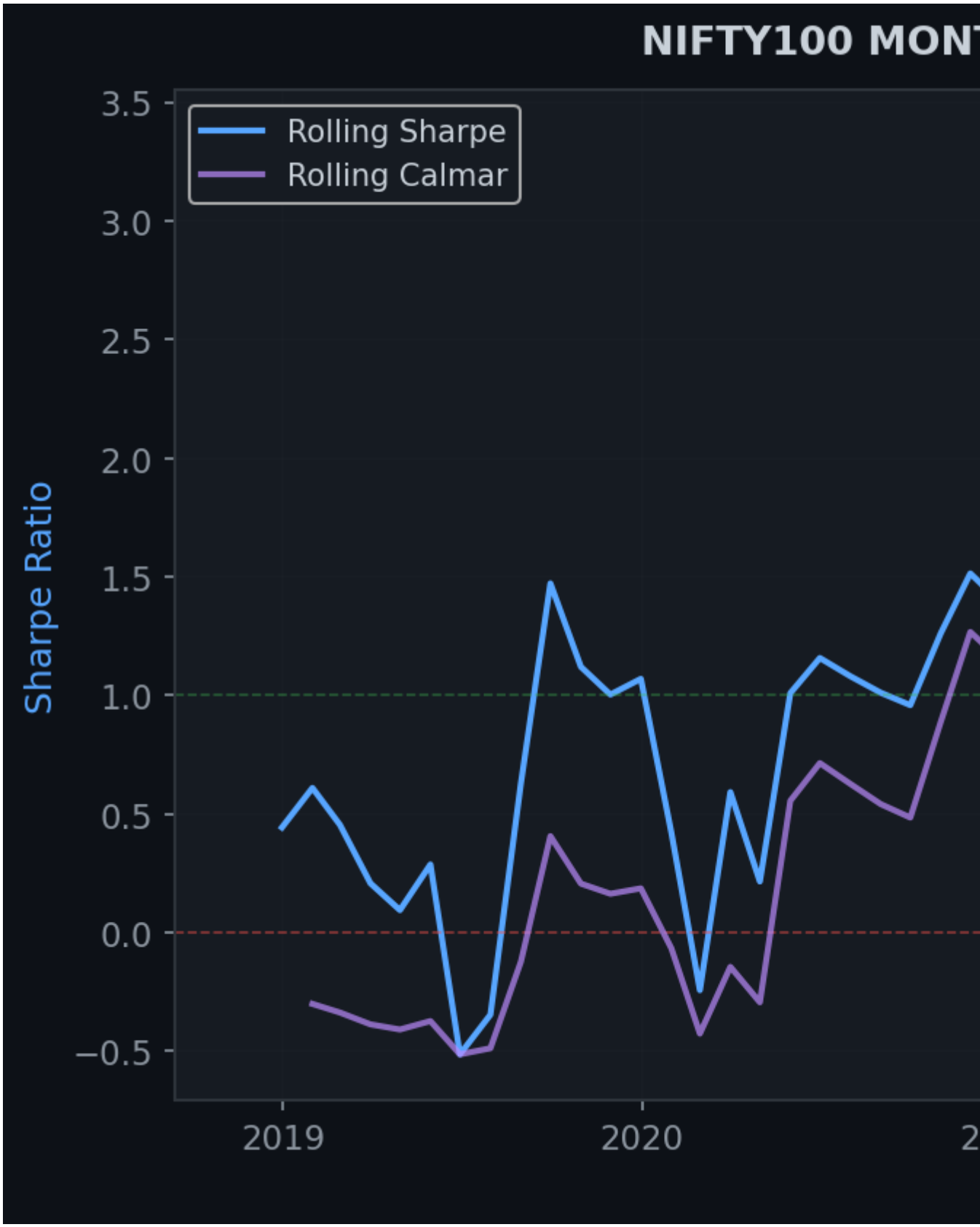
The regime-differentiated stops (âˆ’4/âˆ’6/âˆ’7%) sit at a local optimum. A single flat stop would either over-trigger (too tight) or give up DD control (too loose).

Regime sizing heatmap:


NIFTY100 MONTHLY — Sensitivity: Regime-
(HighVol = MedVol - 1, mi
MedVol Regime — # Holdings
2
3
4
5
6
0.80
1.02
1.04
0.76
0.96
0.97
0.77
0.98
0.99
0.74
0.98
1.01
0.74
1.00
1.04
5
8
10
LowVol Regime — # Holding

The 10/4/3 choice sits at a plateau, not a pinprick. Nearby sizings (9/4/3, 10/5/3) perform similarly â€” the strategy is robust to small perturbations.

Walk-forward stability



Walk-forward stability

Model performance is stable across the expanding training window â€” no sign that early test months enjoyed a softer problem than late ones.

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9. Execution Model

The strategy executes at the **month-end adjusted close** â€” signal formation date and entry/exit price are the same bar. This is the cleanest possible framing: no overnight gap, no first-day-of-month slippage. The per-position trading log (`python3 execution_realism.py --log`) produces a CSV with entry price, exit price, stop-out flag, and realised return for every holding, suitable for manual yfinance verification.

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10. Summary

What makes this work: 1. **PiT composition data** â€” every headline number is survivorship-bias-free. Most retail backtests are not. 2. **Walk-forward discipline everywhere** â€” CatBoost and HMM both retrained only on past data. 3. **Volatility-first regime ordering** â€” stable state labels across refits. 4. `prob Ã— inv-vol` **weighting** â€” rewards conviction, penalises idio risk, wins on every risk-adjusted metric head-to-head. 5. **3M lookback removed** â€” small detail, meaningful Calmar uplift. 6. **Dual-tranche stop-loss** â€” held and new portions tracked independently to avoid false triggers from month-start gap-ups.

What it does not claim: - The 34% CAGR is close-to-close month-end. Execution at the first open of the month will give up some edge; exact slippage depends on execution size and market conditions. - The Sharpe in 2024 narrowed; the soft patch is in the report, not hidden. - Nothing claims â€œAI beats the marketâ€• â€” this is a disciplined factor strategy with a volatility regime overlay. The edge is well-motivated, not mystical.

Headline takeaway: a long-only Indian momentum strategy with 7-year Sharpe of 1.40, peak DD under 11%, every year profitable.

Appendix A â€” Reproduction

```
pip install catboost scikit-learn pandas numpy yfinance scipy hmmlearn mat

# Primary strategy â€” prints headline stats
python3 main.py --index nifty100

# Full validation suite â€— regenerates every chart in output/
python3 validate_strategy.py --index nifty100

# Per-position trading log (CC execution)
python3 execution_realism.py --log

# Post-hoc diagnostics
python3 diagnostics.py --mode annual
python3 diagnostics.py --mode lookback
```

Appendix B â€” Repository

See `README.md` for file-by-file structure and CLI flags.

Appendix C â€” Converting this report to PDF

```
pandoc REPORT.md -o REPORT.pdf --pdf-engine=xelatex -V geometry:margin=1in
```